

Diagrama Base Datos

Diagrama de Base de Datos

Proyecto: QHere - Control de Asistencia Escolar

[Diagrama Mermaid]

erDiagram

schools ||--o{ profiles : asigna
schools ||--o{ grade_sections : contiene
schools ||--o{ schedules : define
schools ||--o{ school_calendar : planifica
schools ||--o{ students : registra
schools ||--o{ authorized_devices : autoriza
schools ||--o{ notification_queue : genera

profiles ||--o{ parents : vincula
profiles ||--o{ student_teachers : imparte
profiles ||--o{ attendance : registra
profiles ||--o{ excuses : revisa
profiles ||--o{ audit_log : ejecuta

grade_sections ||--o{ students : agrupa
students ||--o{ parents : pertenece
students ||--o{ student_teachers : recibe
students ||--o{ attendance : genera
students ||--o{ excuses : justifica
attendance ||--o{ excuses : relaciona
attendance ||--o{ attendance_geo_events : ubica

schools {
 uuid id PK
 text nombre
 text direccion
 text telefono
 text email
 text director
 date academic_period_start
 date academic_period_end
 numeric latitude
 numeric longitude
 numeric allowed_radius_m
 boolean configurado
}

profiles {
 uuid id PK
 uuid school_id FK
 text full_name
 text email
 text role
 text approval_status
 text phone
 text[] permisos
 uuid[] secciones_ids
}

grade_sections {
 uuid id PK
 uuid school_id FK

text grado

text seccion

text turno

boolean special_schedule_enabled

time hora_entrada_especial

time hora_salida_especial

time hora_limite_tardanza_especial

}

schedules {

uuid id PK

uuid school_id FK

text turno

time hora_entrada

time hora_salida

time hora_limite_tardanza

}

school_calendar {

uuid id PK

uuid school_id FK

date fecha

text tipo

text descripcion

}

students {

uuid id PK

uuid school_id FK

uuid grade_section_id FK

text nombre

text matricula

text full_name

text enrollment_code

uuid parent_id FK

text qr_token

boolean activo

}

attendance {

uuid id PK

uuid student_id FK

uuid teacher_id FK

date fecha

time hora_entrada

time hora_salida

text estado

text dispositivo

integer minutos_tarde

}

excuses {

uuid id PK

uuid student_id FK

uuid parent_id FK

uuid attendance_id FK

uuid teacher_id FK

date absence_date

text reason

text attachment_url

text status

}

Notas tecnicas

- `profiles` se relaciona con Supabase Auth por medio de `auth.users(id)`.
- `students.qr\_token` identifica el QR unico de cada estudiante.
- `attendance` registra entrada, salida, tardanza, dispositivo y relacion con docente.
- `audit\_log` conserva trazabilidad de acciones sensibles.
- `authorized\_devices` y `attendance\_geo\_events` cubren control de dispositivo y ubicacion.